CHAPTER 4

A novel secured ledger platform for real-time transactions

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1. Introduction

The invention of blockchain first served as public transaction ledger of the cryptocurrency Bitcoin (Haber and Stornetta, 1991). It came into force through the inventor Satoshi Nakamoto in the year 1991 (Peters et al., 2015). The design of Bitcoin inspired researchers to solve the double-spending problem without the need of a trusted authority or central server. It is a growing list of records that are connected via secured link. Each record, broadly called a block, consists of a hash of the previous block, a time stamp, and transaction data. It is a distributed, decentralized, and public digital ledger useful to keep track of transactions and allows participants to verify and audit transactions independently and inexpensively. A peer-to-peer network and timestamp server manage blockchain database, which is authenticated by mass collaboration powered by collective self-interest. Blockchain finds its application in multiple domains like health-care (Tripathi et al., 2020), data sharing (Shen et al., 2019), supply chain, internet of things (IoT), finance, mobility sharing, etc. (Al-Jaroodi and Mohamed, 2019). Blockchain technology may also be integrated into multiple areas as a distributed ledger for cryptocurrencies, smart contracts, financial services, supply chain, and much more. In Holotescu (2018), the author suggested its usefulness over educational actors and policy makers wanting to explore and to integrate blockchain in institutional projects and curricula.

In Huh et al. (2017), the authors expressed their observations in situations where more than thousands or tens of thousands of IoT devices are connected that the existing model of server-client may have some

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limitations and issues in synchronization. They proposed a method using blockchain to build the IoT system. Here blockchain can control and configure IoT devices. In Watanabe et al. (2015) a proposal was made to use blockchain technology for recording contracts. A new protocol using the technology was described that makes it possible to confirm that contractor consent had been obtained and to archive the contractual document in the blockchain. In another work (Azaria et al., 2016), a modular design integrated with providers' existing, local data storage solutions, facilitating interoperability and making their system convenient and adaptable. They incentivized medical stakeholders (researchers, public health authorities, etc.) to participate in the network as blockchain miners. This provided them with access to aggregate anonymized data with mining rewards in return for sustaining and securing the network via proof of work. MedRec thus enables the emergence of data economics, supplying big data to empower researchers while engaging patients and providers in the choice to release metadata. One article (Turk and Klinc, 2017) incorporates a survey on blockchain technology. It expressed that a construction site blockchain could improve the reliability and trustworthiness of construction logbooks, works performed, and material quantities recorded. In the facility maintenance phase, blockchain's main potential is the secure storage of sensor data, which are sensitive to privacy.

A lightweight blockchain architecture has been proposed (Ismail et al., 2019) for healthcare applications. Other similar work (Hathaliya et al., 2019; Dwivedi et al., 2019) describes a remote patient monitoring application with blockchain and wearable sensors. Other works (Onik et al., 2019; Chen et al., 2019; Su et al., 2020) suggested the use of blockchain in sharing and storing electronic medical records, clinical trial data, and insurance information in the healthcare industry. They proposed a patient-centric data sharing model where the patient is the owner of their own data and chooses to share it with the shareholders as necessary (Al Omar et al., 2019; Leeming et al., 2019). In Abdellatif et al. (2020), edge computing and blockchain were merged together for applications in epidemic discovering, remote monitoring of patients, and fast emergency response. Tele-medical application of blockchain was introduced in Celesti et al. (2020). The authors proposed a tele-medical laboratory service where IoT-based medical devices would be utilized by technicians to perform clinical tests on patients, and the data would be shared over a blockchain and cloud system with the doctors and other hospitals and stakeholders (Ray et al., 2021). Application of blockchain in the healthcare management system has been proposed in other works (Fusco et al., 2020; Jennath et al., 2020). A scalable blockchain architecture with smart contracts was proposed to make IoT networks more secure and better preserve privacy (Satamraju et al., 2020; Abdullah and Jones, 2019; Rathee et al., 2019). As a use case, device authentication, authorization, access-control, and data management for medical purposes were taken. A cross-platform blockchain application having a low energy footprint with multiple stakeholders like patients, hospitals, and pharmaceutical and insurance companies was proposed in Munoz et al. (2019). Their solution followed the EU data regulations for data and privacy protection. A blockchain based tele-surgery framework was proposed in Gupta et al. (2019). They exploited smart contracts to establish trust between all the parties involved.

Combating counterfeit goods is a global problem (Korotkov, 2019). With the onset of blockchain, an effective solution to this problem came to light (Leng et al., 2020). A blockchain framework has been proposed for detecting counterfeit medicine (Saxena et al., 2020; Kumar et al., 2019). Blockchain applications in mobility are gaining momentum in smart cities (Katkova et al., 2020). In Bagloee et al. (2019), a tradable mobility permit based on Bitcoin and Etherium was introduced using tickets for public transportation. Mobility as a service takes into consideration a network of all public transportation systems where only a single permit can be utilized for riding (Nguyen et al., 2019; Vega Medel, 2020). It is possible to achieve a system like this with the help of blockchain (Bothos et al., 2019). A blockchain scheme for group mobility management was proposed (Lai and Ding, 2019). Pollution tracking, car monitoring, and emission control are some important factors involved in car maintenance and registration purposes (Distefano et al.). A blockchain-based architecture to emission monitoring has been proposed in Eckert et al. (2020). Ensuring fuel efficiency of commercial vehicles has been a problem for a long time, and a work (Anwar et al., 2019) tries to tackle it by using blockchain. Smart metering systems in e-mobility is another domain that has been explored with blockchain (Olivares-Rojas et al., 2020).

There have been tremendous advancements in the financial industry involving blockchain (Ali et al., 2020; Baker and Werbach, 2019). It has transcended beyond cryptocurrencies to distributed banking, asset management, and portfolio maintenance (Polyviou et al., 2019; Wang et al., 2017). This change has impacted a lot of business models of different companies as well (Morkunas et al., 2019). Blockchain in agritech is another common area of application (Kamilaris et al., 2019, 2021). Often the consumer is keen to know from where their food comes from, what fertilizers have been utilized to grow it, what is the shelf life of that food, as well as nutrient content, etc. A blockchain-based system is able to give answers to those questions directly (Demestichas et al., 2020). The blockchain-based traceability is able to track food items from inception to consumption (Kamble et al., 2020) (Singh et al., 2020a). In Hang et al. (2020), a fish farm platform was proposed utilizing the concept of blockchain to collect data and maintain data integrity. Blockchain has also found a sweet spot in applications related to law, policy, forensics, and contracts (Temte, 2019; Wanhua, 2020; Dasaklis et al., 2020). In an attempt to involve consumers who produce electricity in the electricity market, a blockchain network was proposed in Diestelmeier (2019). Smart contracts are also being employed in real estate dealings (Seigneur et al., 2020). Blockchain, artificial intelligence, and IoT are transforming smart cities rapidly (Singh et al., 2020b; Gill et al., 2019). In Makhdoom et al. (2020), a privacy-preserving data sharing model was proposed for smart cities. It leverages blockchain to secure public data sharing and enable privacy. Data verification from CCTV cameras of a smart city using IoT and blockchain to prevent crime has been proposed (Khan et al., 2020). A dynamic pricing strategy for energy pricing in a smart city using blockchain helps make smart city energy governance a lot easier (Khattak et al., 2020).

Even though it is so well known, there are some principle drawbacks in the core of blockchain technology, which will be discussed in the further sections. The existing public blockchain technology exhibits a single layer of hashing and incorporates no encryption algorithms for their transactions. Hence, it ascertains partially secured transactions based on one level of hashing to ensure openness to all parties. The property of openness exposes privacy of the users, so we employed a double-layer encryption procedure. The first layer encompasses AES encryption methodology with a randomly generated key at the user end, and the second layer of encryption process is performed at the server end. Data security is ensured through three layers of hashing unlike conventional blockchain networks. The first layer of hashing is initiated at the user end to ensure tamper-proof transmission. The second and third layers are performed at the server side before initiating the validation process to ensure the highest level of data security. Unlike the existing blockchain technology, our proposal ensures real-time transaction and validation even when scaled up. Again unlike conventional blockchain technology, this invention allows multiple applications to perform under single platform. Other than the distinguishing features as mentioned, this invention carries the rest of the features of a blockchain technology. The rest of this paper is organized as follows: Section 2 discusses the brief detailing of the proposed work, followed by Section 3 that analyzes the performance of the system and compares it with some state-of-the-art models. Beside this, Section 4 describes certain use cases of the proposed technology, and the chapter is concluded in Section 5.

2. Proposed work

The work presented in this chapter unveils a novel technology that can be utilized in place of conventional blockchain. In addition to services provided by a normal blockchain technology, it provides the highest security through two layers of encryption, higher authenticity through three layers of hashing utilized for validation, and also gives a platform to trace the complete activity performed by a trusted module over a particular project for future reference. The proposed system does not follow a chain-based file architecture. Due to this, no concept of chain break arises, and the problems that arise as a result of chain break in blockchain are avoided. This system ensures trustworthiness, authenticity, and CIA (confidentiality, integrity, and availability) to its end users while being real time in execution.

The proposed system overview is captured in Fig. 4.1. In the total working environment, there exists a distributed cloud server, a generator, and several validators called trusted modules. These trusted modules continuously communicate with one another while abiding to some

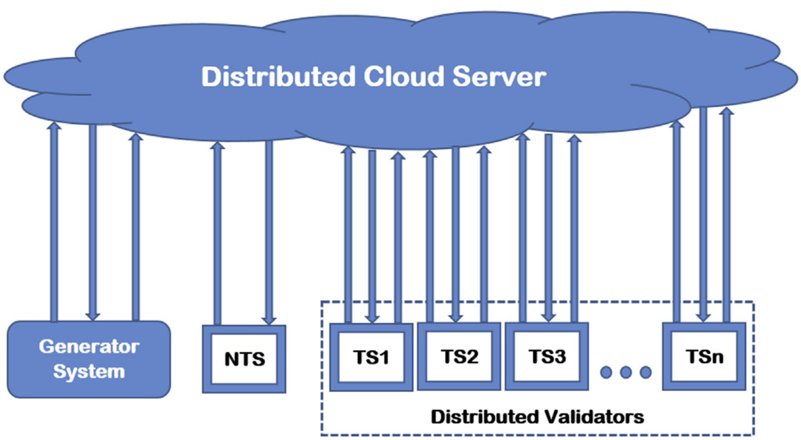

Figure 4.1 System overview.

security rules. If any nontrusted module appears to start communication, the server as a responsible entity shall deny the service requested by the nontrusted module without exposing its data files.

2.1 Communication between generator and server

Fig. 4.2 illustrates the communication between the generator and the server for a single transaction. While generating a combination of data and hash value, namely a node, the generator first sends a request to the server for a random key. After successful authentication, the server generates a random key and sends it to the generator. An encryption process with the data will be carried out at the generator end before it is to be transmitted over the network to the server requesting for further processing.

2.2 Communication between validator and server

Fig. 4.3 depicts communication between the server and validators. A validator, when ready to validate data, modifies its status flag. The server,

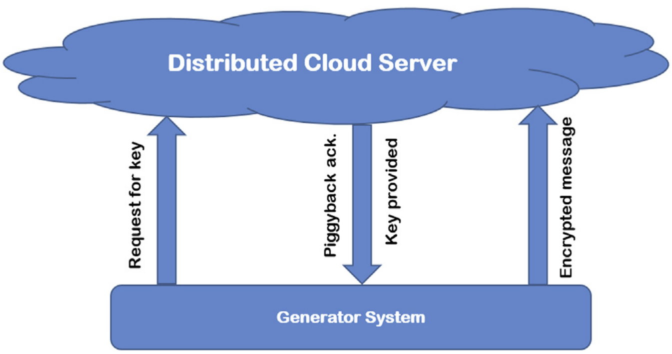

Figure 4.2 Generator-server communication.

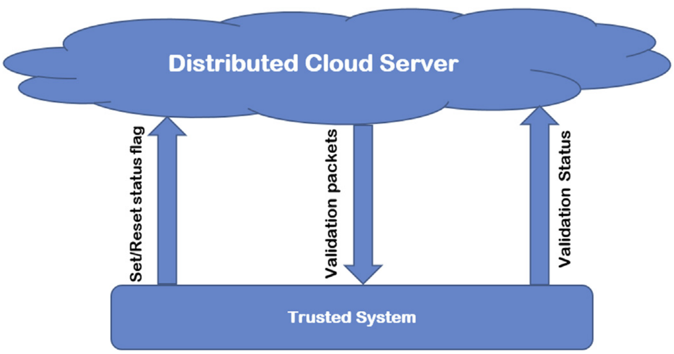

Figure 4.3 Validator-server communication.

when it finds any validator ready, starts sending unvalidated packets toward the validators. The validators on completion of the process of validation send validation status of each packet to the server. While going for a break, validators modify the flag to zero and communicate directly with the server.

2.3 Communication with nontrusted module

Fig. 4.4 depicts a communication attempt by any nontrusted system for gathering protected information from the server. The server plays the role of a moderator and declines any request made by the unauthenticated or nontrusted system. This increases the overall security of the proposed system and prevents unwanted data leakages.

2.4 Functional units of generator module

Fig. 4.5 shows internal data flow of a generator. When a transaction as an event is initiated at the Plain Text Generator, two different operations shall

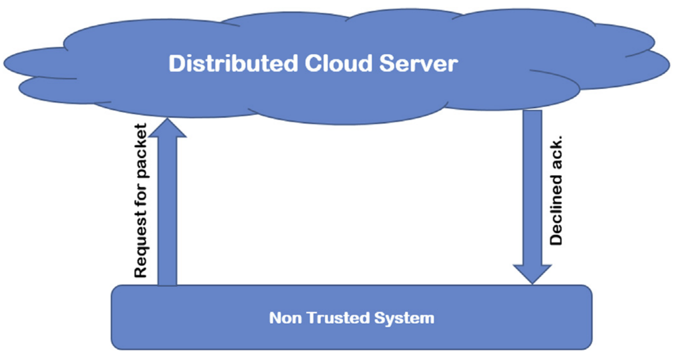

Figure 4.4 Nontrusted module-server communication.

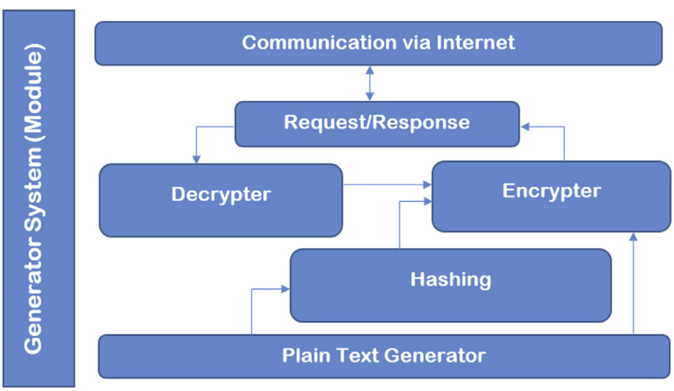

Figure 4.5 Data flow graph for generator.

follow. First and foremost a hash value is generated on data and is forwarded to the Encrypter, which requests a random key from the server. The generated random key is received in encrypted form, which is forwarded to the Decrypter that decrypts the key and communicates the key with the Encrypter. On receiving the key, the encrypter encrypts the combination of plain text and hash value. The Encrypter now communicates the encrypted data to the server.

2.5 Functional units of validator module

Fig. 4.6 illustrates internal data flow for the Validator module. Upon receiving an unvalidated data from the server, it gets stored in the local storage and the validation process gets started. The validation process is completely automated and undergoes four main operations, which are decryption, splitting, hash value computing, and checking for level-wise validation status. These four operations are iterated twice in the same sequence utilizing the local storage during the process. Then the last level of validation is initiated where the data fetched from the local storage is decrypted and split for checking the validation status. This final validation status is communicated to the server. If the validation fails at any stage, the further process is terminated and a negative status is sent to the server.

2.6 Functional units of server module

The internal data flow within the cloud server is demonstrated by Fig. 4.7. At first, the server provides access to the Registration/Login Module to the system accessing the service. Any system that is registered for the service needs to login, and others needs to register per their required service. This

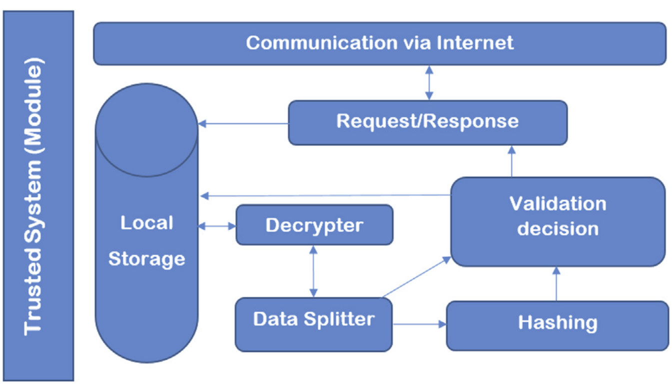

Figure 4.6 Data flow graph for validator.

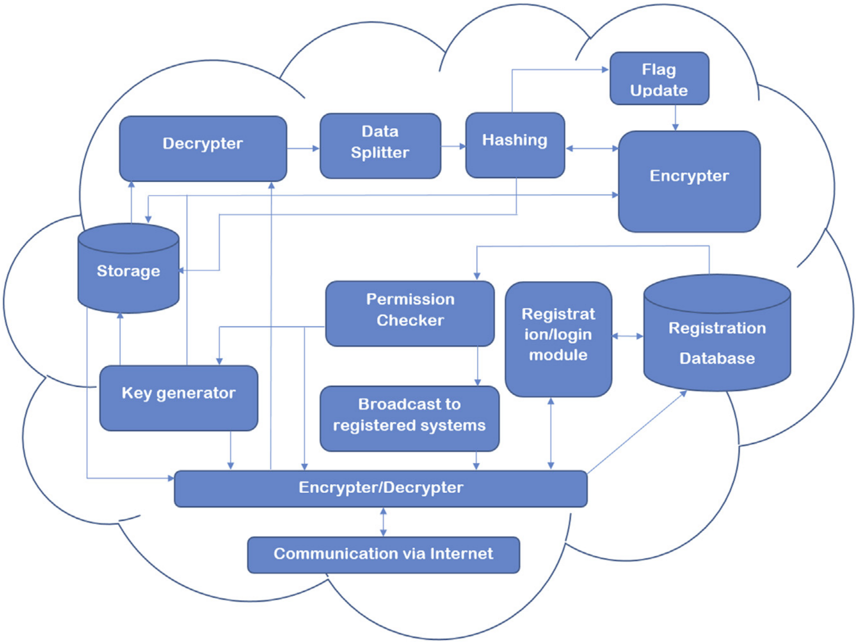

Figure 4.7 Data flow graph for cloud server.

module directly communicates to the Registration Database, which provides the permission details to the Permission Checker, which is further utilized in a later stage. Apart from this service, the server provides the intermediate services to support the proposed platform. This starts when a generator requests a key to initiate a transaction. Upon receiving this request, the server checks the permission from the Permission Checker and ignites the Key Generator module upon successful permission checking. This generated key is transmitted to the generator through a network-level encryption. The cipher text generated by the generator is received by the server and gets forwarded to the local storage. Even a duplicate copy of the cipher text is transmitted to the Decrypter module, which is followed by the Data Splitting module that separates the transaction details from the hash. Now the data is hashed and the hash value is compared with the previous one. If both hash values match, then a status flag is set to 1; otherwise, it is set to 0. This is further followed by the Encrypter module along with hashing modules two times with intermediate interaction with the storage and key generator modules. Then the ultimate details get stored in the Storage module with unvalidated status. This complete process enables the security concept of two-layer encryption and three-layer hashing. Meanwhile, a number of validators are active with validation permission. Now, the unvalidated data are broadcasted to the validators online by the Broadcast to Registered System module upon proper interaction with the Permission Checking module. Once the validators provide feedback, it gets stored. Depending upon the amount of positive and negative feedback from the validators, the ultimate validation status gets updated in the storage module, which basically determines the transaction status.

3. Result and discussion

The proposed system is implemented as an embedded system, and the results show that the system is lightweight and agile. The generator, cloud server, and validators are developed separately and made to communicate with each other over the internet. The system was found to be real time with a time lag of less than 1 s (from generation to validation) including network delay. With a faster network, the delay would become less, which will reduce the overall system delay.

Table 4.1 shows a detailed analysis of the proposed technology when compared with other state-of-the-art ledger-based platforms. With the implementation of dual layers of encryption, the data transmissions are secured to the highest extent. And three layers of hashing ensures data privacy. The proposed methodology does not restrict the number of transactions per second, as it maintains a log book instead of chain structure. Due to adaptation of the log book concept in a ledger platform, several concerns, arising as a result of chain or graph structure, can be avoided. Examples of such concerns include chain break and isolated graph generation due to data hampering.

4. Use cases

1. Healthcare monitoring ecosystem application: We have a patient-centric ecosystem where all medical shops, hospitals, diagnostic centers, patient families, doctors, insurance companies, etc., become parties of the system after they are registered with the application. The application layer should allow different parties to register according to their organizations for proper system maintenance. When a patient registers himself with the application, a permanent patient ID is generated with the help

Table 4.1 Comparison chart.

Methodology	Blockchain (El Ioini and Pahl, 2018)	Sidechain (El Ioini and Pahl, 2018)	Tangle (El Ioini and Pahl, 2018)	Hashgraph (El Ioini and Pahl, 2018)	Proposed
Data structure	Array	List of linked lists	DAG	DAG	Log file
Consensus	PoW: SHA256 hash	PoW: Ethash	PoW: Hashcash	Virtual voting	3 layers of hash
Transactions	Grouped into blocks	Two chains of blocks	Single transaction	Gossip events	Contiguous data
Transactions per second	4 to 7	Limited	500 to 800	>200,000	Unbounded
Validation time	Order of minutes	Order of minutes	Order of seconds	Order of seconds	Order of milliseconds
Privacy	Low	High	Low	Low	Highest
Security	High	High	High	High	Highest

of the disclosed technology. Now, when the treatment starts, the doctor (as a generator) accesses the application layer and communicates with the proposed server through local cloud server and uploads the required documents that need to be sent to the patient (prescription). This ignites the disclosed technology to start its processing, which validates (finds out) any tampering with the document and provides the feedback to the local cloud server. These documents are stored in the patient's application wallet, and selected data from the wallet can be shared with relevant parties according to patient's desire of what data to share, providing application-layer security. Sharing may take place by account number or generator number to indicate which organization should receive the data from the patient via the proposed server. The local server can pull data through API end points of the proposed database (database has transaction instances with generator ID). The moment anything needs to be updated, it takes place in the proposed server upon successful validation. This process continues for subsequent updating, i.e., updates from the doctor, insurance company, medical facility, etc. All communications provided by the proposed server are real-time, secured, and scalable per need.

2. Mobility as a service: We have a traveler-centric ecosystem where all mobility-providing companies become parties of the system after they are registered with the application. The application layer should allow different parties to register according to their organizations for proper system maintenance. When a customer registers with the application, a permanent customer ID is generated with the help of the proposed technology. Now, when the customer initiates a ticket booking with a certain amount of credits, tokens, or stable coins, an initial checking to see if they have the required amount of coins needs to be made. Upon successful checking a transaction needs to be initiated with the proposed technology, and upon receiving successful feedback, a barcode or QR code needs to be generated that would have a certain amount of credits. The payment system of tokens needs to be maintained in the local database where the local database would act as the ledger for the ecosystem. Once the booking process is complete the customer can travel with the barcode or QR code, and each party in the ecosystem will be able to deduct their amount from the credit available in the code. Transferring of credits needs to be again initiated in the application once the QR is scanned by a valid and authentic generator. Credits from the QR code's account will be transferred to the scanning party's

account via the disclosed ledger technology. Upon trip completion the QR will be discarded and relevant data can be stored in the local cloud server for future processing. All communications provided by the proposed server are real-time, secured, and scalable per need.

5. Conclusion

The work revealed here indicates its superiority over existing blockchain technology. It is a platform where multiple transactions of various categories can be performed under one hood. It is a provider of highly secured transaction where three layers of hashing and two layers encryption are incorporated. Three layers of hashing are initiated to prevent tampering of data while communicating between server and generator, within server, and between server and validator. The two layers of encryption manages a man in the middle attack or packet sniffing attack between generator and server and server and validator. The platform provides real-time validation results unlike other existing systems. It also ensures CIA (confidentiality, integrity, and availability) to its end users besides having all other the features except chaining of a blockchain technology.

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